# Smart Inventory Demand Forecasting Decision Support System

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***Abstract*—Inventory planning requires balancing product availability against the financial and operational consequences of excess stock. This paper presents a modular web-based Smart Inventory Demand Forecasting Decision Support System (DSS) that combines supervised machine learning with inventory-risk calculations, scenario analysis, empirical uncertainty information, approximate feature-sensitivity analysis, and a human-in-the-loop approval workflow. The implemented prototype was evaluated using a 48-observation chronological baseline dataset containing four products over twelve months. Product identity was one-hot encoded together with historical sales, inventory, unit price, month, and supplier lead time. Four regression models—Linear Regression, Ridge Regression, Decision Tree, and Random Forest—were evaluated using a chronological 75:25 split, producing 36 training observations and 12 holdout observations. Ridge Regression achieved the lowest MAE of 13.39 and MAPE of 1.70%, while Linear Regression achieved the lowest RMSE of 16.58 and the highest $R^2$ of 0.9935. Ridge was therefore selected according to the predefined MAE-based criterion. The results are treated as preliminary because the holdout contains only 12 observations. The system additionally translates forecasts into safety-stock and reorder-point indicators, supports what-if scenarios, and records managerial approval decisions. The contribution is positioned as a systems-integration and decision-support contribution rather than a new forecasting algorithm. Larger datasets, rolling-origin validation, calibrated uncertainty estimates, and cost-aware inventory optimization are required before production-level claims can be made.**

***Index Terms*—Demand forecasting, inventory management, decision support system, machine learning, Ridge regression, Random Forest, supply chain analytics, inventory optimization.**

## I. INTRODUCTION

Accurate demand estimation is central to inventory planning because purchasing decisions must balance product availability against holding, shortage, and obsolescence consequences. In practical settings, a forecast is useful only when it can be converted into an operational action such as monitoring, replenishment, or escalation. Recent supply-chain research increasingly combines demand forecasting with inventory planning rather than treating forecasting as an isolated analytical task [7], [9], [10], [12].

Machine learning has been applied to retail and supply-chain demand forecasting using structured operational variables and more complex deep-learning architectures. Poongothai *et al.* investigate machine learning for retail demand forecasting and inventory optimization [1], while Wang studies deep-learning-based demand prediction and inventory optimization in e-commerce [2]. Zheng examines machine-learning algorithms for enterprise supply-chain demand forecasting [3], and Singh *et al.* investigate deep learning for forecasting and optimization in e-commerce supply chains [4]. These studies demonstrate the relevance of predictive analytics but use different data settings, objectives, and model families.

The operational connection between forecasting and inventory decisions is particularly important. van der Haar *et al.* study supervised learning for integrated forecasting and inventory control and directly consider feature-driven inventory decisions [12]. Sukolkit *et al.* connect demand forecasting with inventory management in the steel industry [11]. Liu *et al.* investigate machine learning for inventory stockout prediction [14], while An *et al.* address interpretable demand forecasting for micro-fulfillment environments [13]. Lingkon *et al.* further examine multi-item forecasting and its operational effects on replenishment, inventory cost, and service levels [15].

Other application-oriented work demonstrates how forecasting can be connected with broader operational signals. Sathyabama *et al.* combine ARIMA and LSTM with expiry prediction and dynamic pricing for retail inventory management [5]. Manos *et al.* present a dynamic AI-based demand forecasting platform designed to quantify supply-chain value [6]. Nurhasanah *et al.* apply artificial neural networks to demand forecasting within an upstream and midstream supply-chain network [8].

The remaining practical challenge is not the absence of forecasting algorithms. Instead, forecasting outputs must be

connected to an operational workflow in which users can inspect forecasts, understand relevant sensitivities, evaluate inventory risk, perform what-if analysis, and retain control over procurement decisions. This paper therefore presents an integrated DSS rather than proposing a new forecasting algorithm.

The contributions are:

1) a chronological multi-model demand forecasting pipeline;
2) comparison of Linear Regression, Ridge Regression, Decision Tree, and Random Forest;
3) integration of product identity with operational demand features;
4) empirical uncertainty and approximate feature-sensitivity information;
5) deterministic safety-stock and reorder-point calculations;
6) scenario-based what-if analysis; and
7) an auditable human-in-the-loop procurement workflow.

The remainder of this paper is organized as follows. Section II reviews related work. Section III defines the problem and objectives. Section IV presents the system architecture. Section V describes the methodology. Section VI reports the experimental results. Section VII presents implementation and decision-support features. Section VIII discusses limitations and threats to validity. Section IX presents future research directions. Section X concludes the paper.

## II. LITERATURE REVIEW

### A. Demand Forecasting in Supply Chains

Douaioui *et al.* review 119 studies on machine learning and deep learning for supply-chain demand forecasting and identify substantial variation in model selection and application contexts [9]. Goel *et al.* examine both long- and short-term demand forecasting using traditional machine learning and deep learning approaches [10]. Zuvieta *et al.* survey statistical and machine-learning forecasting methods for aviation spare parts, emphasizing the challenges of non-smooth inventory demand [7].

Recent IEEE studies provide application-specific evidence. Poongothai *et al.* study retail demand forecasting and inventory optimization [1]. Wang investigates intelligent demand prediction and inventory optimization in e-commerce [2]. Zheng examines machine-learning algorithms for enterprise supply-chain demand forecasting [3]. Singh *et al.* investigate deep-learning approaches for e-commerce supply-chain forecasting and optimization [4].

### B. Forecasting and Inventory Decisions

Forecast accuracy is most useful when it informs inventory decisions. van der Haar *et al.* explicitly integrate supervised learning with inventory control and show how feature information can be incorporated into order decisions [12]. Sukolkit *et al.* compare forecasting approaches for steel-industry demand and connect forecasting outputs to inventory-management decisions [11]. Lingkon *et al.* propose a hybrid multi-item forecasting method and evaluate downstream effects on replenishment, inventory costs, and service levels [15].

Sathyabama *et al.* combine ARIMA and LSTM with expiry prediction and dynamic pricing for retail inventory management [5]. Manos *et al.* present a dynamic AI-based demand forecasting platform designed to quantify supply-chain value [6]. Nurhasanah *et al.* demonstrate neural-network demand forecasting in a multi-stage supply-chain setting [8]. These studies support the need to connect forecasts with operational decisions while demonstrating that different domains require different modeling assumptions.

### C. Inventory Risk and Interpretability

Inventory risk is not limited to forecast error. Liu *et al.* study machine learning for inventory stockout prediction, highlighting the importance of predictive risk indicators in inventory management [14]. An *et al.* develop association-rule-based machine learning for demand forecasting in micro-fulfillment centers and emphasize both forecasting performance and interpretability [13]. These findings motivate the inclusion of approximate feature sensitivity and inventory-risk indicators in the proposed DSS.

### D. Research Gap

The literature establishes strong evidence for machine-learning demand forecasting, inventory optimization, stockout prediction, and interpretable forecasting. However, these components are often investigated as separate analytical problems or within specialized operational settings. The gap addressed by this work is therefore an integration gap: the prototype connects forecasting, inventory-risk calculations, scenario analysis, approximate explanations, and managerial approval in one auditable workflow. The contribution is consequently positioned as a systems-integration and decision-support contribution rather than algorithmic novelty.

## III. PROBLEM STATEMENT AND OBJECTIVES

The proposed system addresses the operational problem of translating uncertain future demand into transparent inventory decisions. A forecasting module alone cannot determine whether an item should be reordered because the action depends on current inventory, supplier lead time, service-level assumptions, and the uncertainty surrounding demand.

Let $D_t$ denote demand in period $t$, $X_t$ the information available at the prediction origin, and $\hat{D}_{t+1} = f(X_t)$ the forecast. The decision-support problem can be expressed as:

$$\hat{D}_{t+1} = f(X_t), \qquad A_{t+1} = g(\hat{D}_{t+1}, I_t, L_t, U_t), \quad (1)$$

where $I_t$ is inventory position, $L_t$ is lead time, $U_t$ represents uncertainty-related information, and $A_{t+1}$ is a recommended action. The function $g(\cdot)$ is intentionally deterministic in the current prototype so that a user can inspect the assumptions behind a recommendation.

The objectives are:

1) develop a modular web-based inventory DSS;

TABLE I
SYNTHESIS OF RECENT RESEARCH RELATED TO THE PROPOSED SYSTEM

| Reference | Focus | Key relevance | Relation to this work |
|---|---|---|---|
| [9] | ML/DL supply-chain forecasting review | Broad comparison of forecasting approaches | Literature foundation |
| [1] | Retail forecasting and inventory optimization | ML for retail inventory decisions | Retail context |
| [2] | E-commerce demand prediction | Deep learning with inventory optimization | Advanced-model direction |
| [3] | Enterprise demand forecasting | ML model application | Comparative modeling |
| [4] | E-commerce forecasting and optimization | Deep-learning supply-chain application | Future model extension |
| [12] | Integrated forecasting and inventory control | Direct forecasting-to-inventory integration | Core DSS motivation |
| [11] | Inventory-oriented demand forecasting | Forecasting connected to inventory management | Inventory integration |
| [13] | Interpretable MFC forecasting | ML plus interpretability | Explainability motivation |
| [14] | Inventory stockout prediction | ML-based inventory risk | Risk-layer motivation |
| [15] | Multi-item supply-chain forecasting | Replenishment, inventory cost, service level | Multi-product direction |

2) compare four supervised regression models using chronological validation;
3) preserve a time-ordered relationship between predictor information and target demand;
4) integrate product identity and operational features;
5) provide empirical forecast uncertainty and error metrics;
6) calculate safety stock and reorder point;
7) provide what-if scenario analysis; and
8) retain human authority through an auditable approval workflow.

## IV. SYSTEM ARCHITECTURE

The system follows a layered architecture that separates user interaction, application services, predictive analytics, decision logic, and persistence. This separation reduces coupling between the web application and the forecasting implementation and allows individual layers to be replaced in future iterations.

### A. Presentation Layer

The presentation layer provides a consolidated interface for product selection, inventory status, demand forecasts, scenario analysis, and recommendation review. The design separates informational views from action controls so that a forecast is not automatically interpreted as an order instruction.

### B. Application and Authentication Layer

The Node.js and Express backend exposes REST-style application endpoints. Authentication is handled with JSON Web Tokens (JWTs), while protected operations are separated from public application resources. The backend coordinates data retrieval, feature preparation, model invocation, decision calculation, and audit recording.

### C. Forecasting and Analytics Layer

The forecasting service receives a validated feature vector and invokes the selected serialized scikit-learn model. The analytics layer computes evaluation metrics during experimentation and produces empirical uncertainty and approximate feature-sensitivity information for supported models.

### D. Inventory Decision Layer

The decision engine translates the demand forecast into daily demand, safety stock, reorder point, and a prototype order quantity. The layer is intentionally deterministic so that each recommendation can be traced to explicit assumptions.

### E. Persistence and Audit Layer

The development prototype uses JSON persistence for product, inventory, supplier, forecast, and audit records. Although this approach is not intended to demonstrate production scalability, it makes the experimental workflow lightweight and reproducible. The audit record provides a historical trail of recommendation actions and user justification.

## V. METHODOLOGY

### A. Data Description

The baseline dataset contains 48 chronological observations representing four products over twelve months. The products are Laptop Pro X1, Wireless Headphones, Smart Watch S3, and Gaming Mouse. Each observation contains product identity, previous-period sales, inventory position, unit price, month, supplier lead time, and target-period demand.

Product identity is one-hot encoded. The baseline feature vector is:

$$X_t = [P_t, S_{t-1}, I_t, Prc_t, M_t, L_t]. \tag{2}$$

The target is:

$$y_t = Demand_t. \tag{3}$$

The use of $Sales_{t-1}$ explicitly represents information available before the target period. The raw dataset and training code should be retained with the submission materials so that the temporal construction can be independently reproduced.

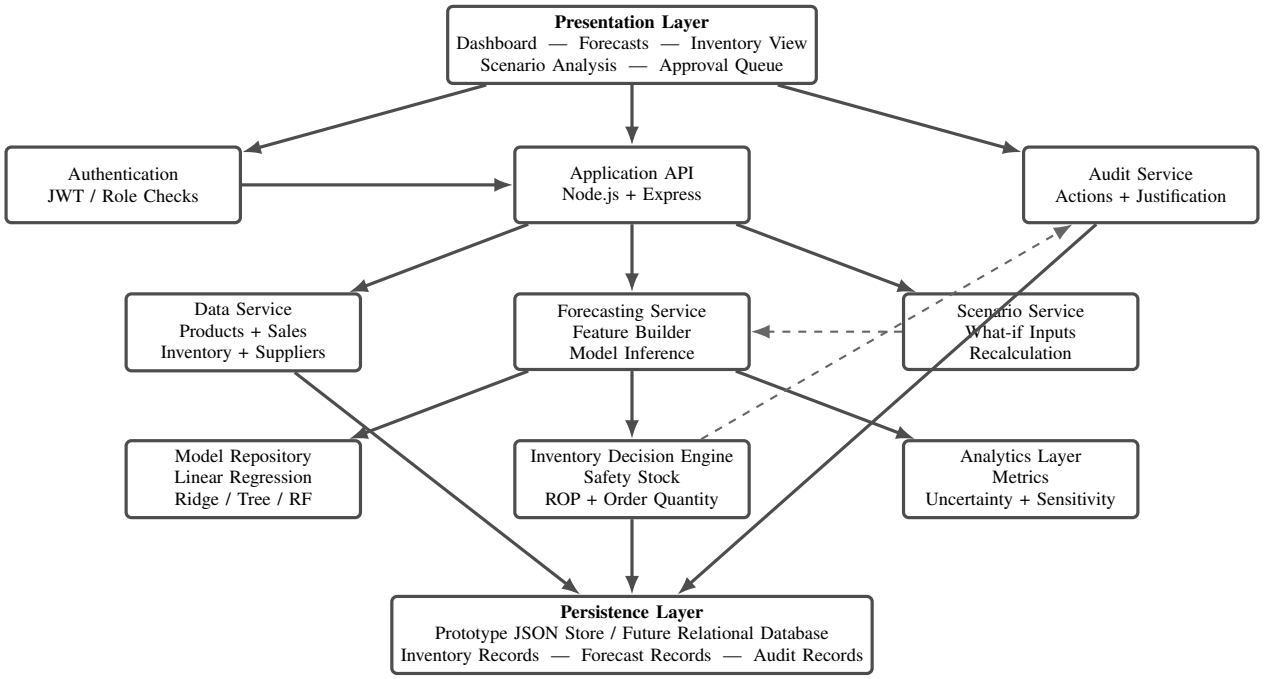



Fig. 1. Layered architecture of the Smart Inventory Demand Forecasting Decision Support System. Solid arrows represent primary application or data flow; dashed arrows represent feedback or audit interactions.

TABLE II
BASELINE DATASET AND FEATURE ROLES

| Variable | Role in forecasting |
|---|---|
| Product ID | One-hot encoded product identity |
| Previous sales | Lagged demand signal available before target period |
| Inventory | Inventory position at prediction origin |
| Unit price | Operational pricing signal |
| Month | Calendar-period indicator |
| Lead time | Supplier delivery-time information |
| Demand | Target variable |

### *B. Data Quality and Preprocessing*

The preprocessing stage checks schema consistency, missing values, duplicates, and numerical observations. Potential outliers are identified using the interquartile range:

$$IQR = Q_3 - Q_1. \tag{4}$$

Observations below $Q_1 - 1.5IQR$ or above $Q_3 + 1.5IQR$ are flagged. Twelve price observations were flagged in the supplied baseline data and retained for the reported experiment because they were treated as legitimate observations rather than automatically deleted.

Continuous variables used by Linear Regression and Ridge Regression are standardized using training-set statistics:

$$x' = \frac{x - \mu_{train}}{\sigma_{train}}. \tag{5}$$

The same parameters are applied to holdout observations. Tree-based models do not require scaling.

### *C. Temporal Split*

The first nine monthly observations for each product are used for training and the final three for testing:

$$N_{train} = 4 \times 9 = 36, \qquad N_{test} = 4 \times 3 = 12. \tag{6}$$

No random shuffling is used. The chronological split reflects the intended deployment direction, in which historical information is used to predict later observations.

### *D. Forecasting Models*

Four supervised regression models are evaluated. Linear Regression provides a transparent baseline. Ridge adds L2 regularization to reduce coefficient sensitivity. Decision Tree captures nonlinear threshold relationships, while Random Forest aggregates multiple trees to reduce the variance of an individual tree.

Linear Regression minimizes:

$$\min_w ||Xw - y||_2^2. \tag{7}$$

Ridge Regression minimizes:

$$\min_w ||Xw - y||_2^2 + \alpha||w||_2^2, \tag{8}$$

with $\alpha = 1.0$.

The Decision Tree uses maximum depth five. The Random Forest uses 100 trees. The configurations are summarized in Table III.

TABLE III
MODEL CONFIGURATIONS

| Model | Configuration |
|---|---|
| Linear Regression | fit intercept |
| Ridge Regression | $\alpha = 1.0$ |
| Decision Tree | maximum depth = 5 |
| Random Forest | 100 trees |

### E. Evaluation Metrics

The models are evaluated using MAE, RMSE, MAPE, and $R^2$:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|, \tag{9}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}, \tag{10}$$

$$MAPE = \frac{100}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right|, \tag{11}$$

and

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}. \tag{12}$$

MAE is the primary model-selection criterion because it expresses forecast error directly in demand units. RMSE is retained to expose larger errors, while MAPE and $R^2$ provide complementary views.

### F. Uncertainty and Feature Sensitivity

For Random Forest, the standard deviation of individual tree predictions is used as an empirical dispersion estimate:

$$\sigma_{trees} = Std(f_1(x), \ldots, f_K(x)). \tag{13}$$

An empirical normal-approximation band is:

$$U_{90} = \hat{y} \pm 1.645\sigma_{trees}. \tag{14}$$

This is not claimed to be a calibrated 90% prediction interval.

For Ridge, uncertainty is estimated from chronological training residuals. Feature sensitivity is estimated using baseline perturbation: one feature is changed to its active value while the remaining features are held at a baseline vector, and the resulting prediction difference is recorded. This is an approximate model-specific sensitivity measure and is not claimed to provide formal SHAP attribution.

### G. Inventory Decision Formulation

The forecast is passed to a deterministic inventory layer. Approximate daily demand is:

$$\bar{d} = \frac{\hat{D}}{30}. \tag{15}$$

The generalized safety-stock expression is:

$$SS = Z\sqrt{\bar{L}\sigma_d^2 + \bar{d}^2\sigma_L^2}. \tag{16}$$

With negligible lead-time variability:

$$SS = Z\sigma_d\sqrt{\bar{L}}. \tag{17}$$

The prototype uses $Z = 1.65$. The reorder point is:

$$ROP = \bar{d}\bar{L} + SS. \tag{18}$$

The prototype order quantity is:

$$Q = \max(0, \hat{D} + SS - I), \tag{19}$$

where $I$ is current inventory. This is a transparent replenishment heuristic rather than a cost-optimal inventory-control policy.

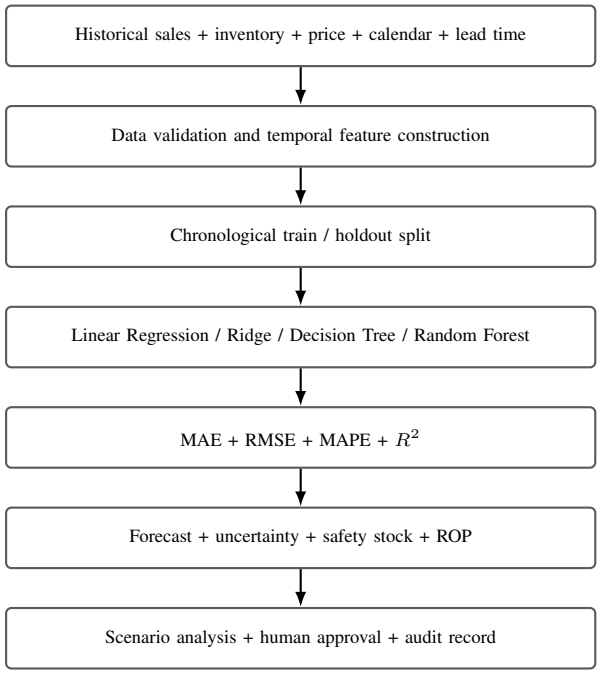



Fig. 2. End-to-end analytical workflow from historical observations to an auditable inventory decision.

## VI. EXPERIMENTAL RESULTS

The experiment uses the 48-row baseline dataset and the 12-observation chronological holdout. The software environment is Node.js 18.x, Python 3.10, pandas 2.0, and scikit-learn 1.2.

TABLE IV
MODEL PERFORMANCE ON THE 12-OBSERVATION HOLDOUT

| Model | MAE | RMSE | MAPE | $R^2$ |
|---|---|---|---|---|
| Random Forest | 48.10 | 60.30 | 5.97% | 0.9140 |
| Linear Regression | 13.87 | **16.58** | 2.00% | **0.9935** |
| **Ridge Regression** | **13.39** | 17.70 | **1.70%** | 0.9926 |
| Decision Tree | 44.72 | 54.34 | 5.58% | 0.9304 |

Ridge achieves the lowest MAE and MAPE. Linear Regression achieves the lowest RMSE and highest $R^2$. Therefore, Ridge is selected according to the predefined MAE criterion; it is not claimed to dominate every metric.

The difference between Ridge and Linear Regression in MAE is small, while the difference in RMSE favors Linear Regression. This indicates that model selection depends on the operational loss function. For an inventory setting in which the average magnitude of error is the primary concern, MAE provides an interpretable selection rule. If unusually large errors carry substantially higher cost, RMSE or an asymmetric cost function would become more appropriate.

The Random Forest and Decision Tree results are considerably weaker on the holdout. This does not imply that tree-based methods are generally inferior. Instead, it suggests that the small baseline dataset may not contain enough observations to support reliable nonlinear partitioning. On larger datasets, hyperparameter tuning and richer temporal features could change the relative ranking.

### A. Interpretation of High $R^2$

The linear models produce $R^2$ values above 0.99. Such values are unusually high for a small forecasting experiment. Possible explanations include strong structure in the supplied dataset, highly informative lagged features, or leakage in data construction. The paper therefore does not generalize these values to larger commercial datasets. The raw dataset and training code should be retained so that the temporal construction and reported metrics can be independently reproduced.

### B. Operational Interpretation

The forecasting result is not itself a procurement order. A forecast is passed to the inventory layer, where current stock, lead time, service-level assumptions, and uncertainty are considered. This separation is important because the same forecast can lead to different actions depending on the current inventory position. For example, a high forecast with substantial available stock may lead to monitoring rather than immediate ordering, whereas the same forecast with low stock and long lead time may trigger an elevated replenishment recommendation.

## VII. IMPLEMENTATION AND DECISION SUPPORT

### A. Scenario Analysis

The scenario module allows users to modify selected assumptions such as price, lead time, inventory, and demand-related inputs before accepting a forecast. The modified values are evaluated without permanently altering historical records. Scenario outputs therefore function as decision-support evidence rather than as changes to the underlying training dataset.

### B. Recommendation Workflow

The recommendation workflow separates prediction from authorization. A forecast can generate a recommendation, but the recommendation enters a pending state before a user action is recorded. The operator can choose APPROVE, REJECT, or SNOOZE. For rejected or delayed recommendations, the user can provide a contextual reason. The audit record stores the timestamp, user identifier, selected model, forecast, recommendation, action, and justification.

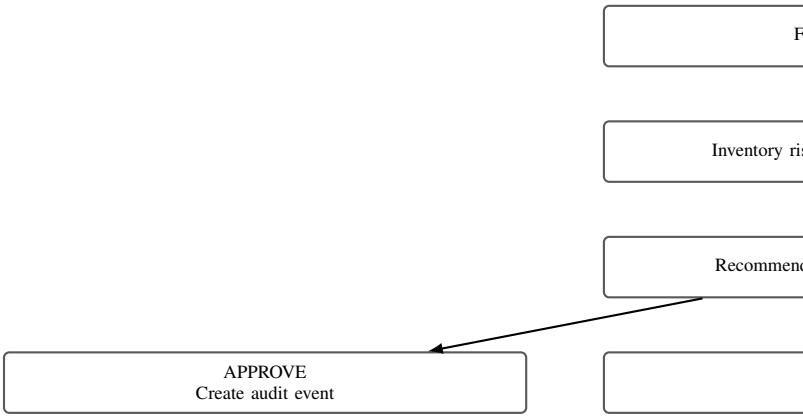



Fig. 3. Human-in-the-loop recommendation workflow.

### C. Security and Audit Considerations

JWT-based authentication restricts protected operations to authenticated users. In addition to access control, the audit layer supports accountability by retaining recommendation outcomes and justifications. The current prototype does not claim enterprise-grade security testing; security hardening, secret management, rate limiting, and centralized logging are deployment requirements for future production versions.

### D. Reproducibility

Reproducibility depends on preserving the dataset, preprocessing rules, model configurations, training split, and evaluation scripts. The reported experiment uses fixed model configurations and a deterministic chronological split. Future releases should additionally record package versions, random seeds for stochastic models, model hashes, and dataset versions.

## VIII. LIMITATIONS AND THREATS TO VALIDITY

The baseline dataset contains only 48 observations and four products over twelve months. The final evaluation contains only 12 observations. Consequently, the experiment cannot reliably establish long-term seasonality, holiday effects, structural changes, or broad market generalization.

The evaluation uses one chronological holdout rather than rolling-origin validation. Product identity is encoded, but the small number of observations limits the strength of product-specific inference. The localized JSON persistence layer is suitable for development but does not establish production-scale database performance. The system does not explicitly model censored demand caused by stockouts. The empirical uncertainty estimate is not statistically calibrated, and baseline perturbation does not provide formal SHAP attribution guarantees.

The inventory layer also simplifies lead-time variability and uses a deterministic replenishment heuristic. A complete inventory-optimization study would require explicit holding, shortage, ordering, and service-level costs. Therefore, the current paper evaluates a decision-support prototype rather than claiming optimal inventory control.

There is also a risk of dataset-specific overfitting. The unusually high linear-model $R^2$ values motivate additional validation before deployment. Rolling-origin evaluation over multiple forecast origins, external datasets, and a larger number of SKUs would provide stronger evidence.

## IX. FUTURE RESEARCH DIRECTIONS

Future work should evaluate larger multi-year datasets with more SKUs, promotions, holidays, stockout indicators, and external demand drivers. Rolling-origin or walk-forward validation should replace the single holdout for stronger temporal evaluation. Advanced models such as gradient boosting, LSTM, GRU, or transformer-based forecasting can then be compared when sufficient data are available.

The inventory layer can be extended from deterministic prototype rules to cost-aware optimization involving holding cost, ordering cost, shortage cost, service-level constraints, and lead-time uncertainty. Uncertainty estimation can be evaluated using calibrated probabilistic forecasting or conformal prediction. The explainability layer can be compared against formal methods such as SHAP after the dataset size is increased.

The prototype persistence layer can be migrated to a relational database and evaluated under realistic multi-user workloads. A future deployment can also introduce scheduled model retraining, model versioning, data-drift monitoring, automated data-quality checks, and role-based approval policies.

Finally, a stronger evaluation should measure not only forecast error but also operational outcomes such as stockout frequency, average inventory, service level, inventory turnover, and total inventory cost. This would connect the predictive metrics to the actual business objective of the system.

## X. CONCLUSION

This paper presented a Smart Inventory Demand Forecasting Decision Support System integrating supervised demand forecasting, inventory decision logic, scenario analysis, empirical uncertainty, approximate feature sensitivity, and human approval. On the supplied 48-row baseline dataset, Ridge Regression obtained the lowest MAE of 13.39 and MAPE of 1.70%, while Linear Regression obtained a slightly lower RMSE and higher $R^2$.

These results are preliminary because only twelve observations form the chronological holdout. The primary contribution is therefore the integration of forecasting and inventory decision support into an auditable workflow rather than a claim of a new forecasting algorithm. The architecture separates prediction from operational decision logic and preserves human authority over procurement actions.

Larger datasets, stronger temporal validation, calibrated uncertainty, explicit inventory-cost objectives, and operational outcome evaluation are required before production-level claims can be made. The proposed prototype nevertheless provides a structured foundation for extending demand forecasting into a transparent and auditable inventory decision-support workflow.

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